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## First Solar: CFRA's Accounting Quality Concerns

On May 2, 2009 analysts at the Center for Financial Research and Analysis (CFRA)<sup>a</sup>, reviewed a copy of First Solar's May 1<sup>st</sup> first quarter 10-Q. It was up to CFRA's alternative energy team to quickly digest the most recent earnings data from the leading U.S.-based integrated solar company and produce a report clients were eagerly awaiting. CFRA had only recently initiated coverage of the solar industry, publishing its first industry report in January 2008. The report, which examined the accounting practices of leading firms and questioned the continued meteoric climb of the industry, cautioned clients to be skeptical of financial statements, some of which CFRA considered to be products of aggressive accounting. With the onset of the global financial crisis, investors were increasingly jittery and impatient for CFRA's analysis.

CFRA was founded in 1994 as a consulting firm working with mutual funds to research accounting practices of companies in their portfolios. By 2008, CFRA had become a leader in forensic accounting specializing in investigating aggressive accounting practices and in extreme cases fraud. CFRA clients included roughly 40% hedge funds, 40% mutual funds, and a 20% mix of insurance companies and law firms. The global research team consisted of 25 analysts covering six industries in 25 countries spanning Europe and North America.<sup>b</sup> The analysts, who averaged over 10 years of accounting and financial experience, were a mix of MBAs, CPAs and CFAs.

Before diving into the new 10-Q the analysts wanted to reacquaint themselves with the industry and areas of potential risk. The initial industry report had examined revenue quality and recognition, deferred revenues, inventory levels, customer and geographic diversification, warranty policies and accruals, production capacity growth and supply chain risks. Despite the many red flags in the report, First Solar was rarely mentioned as a concern. However, with the financial crisis setting in, the team knew that a great deal had changed in a short period of time.

### CFRA's Products and Methods

CFRA's core products were research reports written on individual companies suspected of aggressive accounting practices that resulted in inflated earnings which presented a healthier

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<sup>a</sup> CFRA was a subsidiary of Risk Metrics. In some instances Risk Metrics is referenced in the text, the names are used interchangeably.

<sup>b</sup> Covered industries included financials, consumer, healthcare, industrials, technology-media-telecom and energy.

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financial outlook than warranted. The result of CFRA investigations was the publication and upkeep of a “biggest concerns” list, or a list of companies CFRA felt were due to lose value when the benefits of their aggressive accounting practices ran out. The list contained roughly 20 to 35 companies from the U.S. and another 10 to 15 from Europe.<sup>1</sup>

Dan Mahoney, CFRA's director of research, explained the process of placing a company on the biggest concern list:

There are a series of questions we ask. The first question is, “does the market understand what is going on [in the financial disclosures]?” If it does, it is probably not a biggest concern. If the market does not understand, the next question is: “will the behavior impact the operations, or financial results, enough to move this stock in a negative way?” Again, if it won't, then we move on. However, if it will, then we ask, ‘did management address the concern?’ If it's not understood by the market, it is material and management's explanations don't satisfactorily mitigate our concern that probably means it warrants being a biggest concern.<sup>2</sup>

The CFRA research approach looked specifically at earnings quality and accounting issues presented in company SEC and other public filings. Mahoney explained the methodology, “We are broken into industry teams. We run proprietary screens internally that look for working capital issues like day's sales in inventories, day's sales in receivables, accruals, and cash flow versus net income.” He continued with an example,

Let's assume you have an industry of 400 companies. Every quarter, we'll screen them and get a subset of 30 that warrant additional digging. The next step is to dig into the filings, conference calls, company presentations, anything we can get our hands on and try to figure out whether the red flag is an actual problem or an anomaly in the data. Then we build a thesis around the problem. What we really try to avoid is being a red flag service. We don't publish something on a company highlighting a bunch of red flags; we try to develop a specific thesis about the company.<sup>3</sup>

Additionally, CFRA would screen companies in certain high risk industries, such as new technologies, or let analysts follow their intuitions on industries and particular companies. In some cases CFRA compared filings across quarters to detect changes. Specifically, analysts utilized word searches to identify key phrases that might indicate the adoption of aggressive accounting techniques. Mahoney clarified,

We look for specific language in a filing that will indicate that the accounting has been changed. For example we look for the phrases, ‘extended the useful life,’ ‘extended credit terms,’ or ‘extended payment terms.’ If we can find these words in the filing, it may indicate that the company has extended the useful life of assets for depreciation purposes or extended credit terms to drive revenue. So there are a bunch of these different searches that we do to identify changes in accounting policy or changes in business practice.<sup>4</sup>

In addition to its core research, CFRA offered clients numerous secondary services including individualized coverage of companies not included on the biggest concerns list, legal analysis of pending litigation that related to public companies, seminars and tutorials on accounting issues and industry briefings. They also developed and maintained QuickScore, a collection of quantitative analysis that evaluated over 10,000 companies on earnings and cash flow risk.

## The Solar Industry

In a given 24-hour period the sunlight striking the earth's surface generated enough power to satisfy the energy needs of each 6.6 billion inhabitant for a period of 27 years.<sup>5</sup> This immense potential made solar the most abundant form of alternative energy available on the planet. The vast possibilities combined with growing concerns over global warming, the ecological impact of fossil fuel extraction, rising energy demand and energy independence led many to champion the future of solar power. However, the technical challenges of harnessing the sun's energy combined with the financing challenges of building power plants and storage facilities undercut scalability and prospects for universal implementation.

By 2009, despite the challenges, the industry, as measured by the production of solar panels, had grown 48% a year worldwide since 2002.<sup>6</sup> In the U.S., during the same timeframe, the solar market grew from 168 megawatts (MW) to 1,111 MW. The U.S.-based growth was tied to technological advancement and a subsequent decrease in price per watt of energy produced. In 1998 each watt installed cost \$10.50, by 2007 the price had dropped to \$7.60, or roughly a 3.5% annual decrease in real dollars.<sup>7</sup> Despite these cost reductions, solar remained one of the most expensive forms of energy generation. Globally, solar was five-times more expensive than coal, two-and-a-half times more expensive than wind and twice as expensive as nuclear.<sup>8</sup>

To offset the higher costs and encourage investment in the sector, governments provided subsidies and tax incentives which further grew the industry. In the U.S., the government offered a 30% capital subsidy through the Investment Tax Credit (ITC) for renewable power, including solar. However, as many early solar companies had limited profits many were not eligible for the subsidy. In response, in early 2009, the U.S. Treasury began offering cash grants and loan guarantees to solar energy producers. European governments relied on feed-in tariffs (FIT) to encourage private investment in the solar industry. A FIT required power utilities to buy renewable energy (wind, solar, etc.) at above market prices from providers. The guaranteed consumer base encouraged companies to invest in solar and PV grids. In return, the utilities passed the extra cost onto consumers through small surcharges. FIT were particularly popular in Spain, Germany and Italy who accounted for roughly 45%, 25% and 3% of the global PV module market in 2008, respectively. The U.S. accounted for an additional 5% and Japan 3%.<sup>9</sup>

### *The Solar Value Chain*

Photovoltaic systems (PV), or solar panels, were largely used for two purposes. The majority, or roughly 90% converted solar rays into electricity which could then be connected to a single appliance, building or electrical grid. The remaining 10% of panels were engaged in thermal systems which used the sun's energy to heat water for home water heaters or, on a larger scale, steam generation that powered electrical turbines.<sup>10</sup> The solar value chain was broken into three distinct phases:

**Upstream** The upstream market consisted of polysilicon production and the fabrication of solar wafers. Solar wafers, or cells, were the general building block of solar modules, and thus solar power systems. In the solar value chain, the manufacture of solar wafers was highly capital intensive but also captured the largest percentage of profits.<sup>11</sup> Building a polysilicon plant could cost between \$500 million and \$1 billion, a significant barrier to entry.<sup>12</sup>

For 30 years the majority of the solar industry had produced PV panels with crystalline silicon, which despite higher production expenditures, sought cost effectiveness through high-efficiency energy conversion.<sup>13</sup> The majority of these panels maintained conversion rates over 20%.<sup>14</sup> The

alternative was cadmium telluride which had lower production costs but also a lower conversion rate of just under 10%.<sup>15</sup> Manufactured as a thin film, cadmium telluride was light and could be stored and shipped easily reducing costs. Conversely, crystalline silicon was only manufactured as square panels that weighed more and were not malleable. Cadmium telluride required 0.22 grams to produce one Watt of power while crystalline polysilicon required 8 grams.<sup>16</sup>

**Midstream** The midstream market, the main segment of the solar industry, was converting silicon cells into solar modules, the panels on the roofs of houses or in larger solar fields.<sup>17</sup> Modules typically weighed 30 to 60 lbs. and included 60 to 72 cells covering an area of roughly one square yard.<sup>18</sup> Barriers to entry at the midstream level were quite low as proprietary manufacturing processes were limited and replication was uncomplicated.<sup>19</sup> At this stage profitability depended on lowering production costs.<sup>20</sup>

**Downstream** The downstream market consisted of the design, construction, management and maintenance of solar power systems. Large systems could consist of anywhere from a hundred to tens-of-thousands of solar modules lined in an open space with direct sun exposure. Large solar projects were often commissioned by public utilities or large private companies with agreements to provide energy to existing electrical grids. In vertically integrated solar companies, downstream service were referred as the Balance of Systems (BoS), or the costs associated with the purchase and construction of mounting structures, electrical components including inverters, AC switches, junction boxes and transformers, plus all administrative costs including labor, financing, permitting and relations with the community, state and utility.<sup>21</sup>

## First Solar

First Solar's early roots were in the glass business. In 1948, Harold McMaster made his fortune producing permaglass, tempered glass that shattered into hundreds of small shards as opposed to dangerously sharp jagged pieces, for the growing US automotive industry.<sup>22</sup> In 1984, McMaster turned his attention to the solar industry. In an attempt to achieve cost competitiveness he began experimenting with alternatives to the industry accepted crystalline silicon approach, eventually landing on cheaper and lighter cadmium telluride. After numerous unsuccessful solar ventures, including infusions of cash from himself and outside investors, First Solar came into being in 1999.<sup>23</sup>

Having settled on the technology, First Solar began mass producing solar cells at an Ohio-based plant in 2005. As the producer of solar cells, First Solar found success in Europe, particularly Germany, where the overall solar ecosystem, including inverters, wiring and site construction, was better developed. Conversely, in the U.S. it was difficult to sell panels without offering clients full project support. In an attempt to reach the U.S. market, First Solar began vertically integrating through acquisition. In 2007, following their 2006 IPO, the company acquired Turner Renewable Energy, which had experience in the U.S. market developing independent power projects, marketing energy, and operating commercial power plants.<sup>24</sup> In early 2009, prior to the publication of its Q1 10-Q, First Solar announced the purchase of California-based OptiSolar, a rival company with a significant project pipeline. The acquisition allowed First Solar panels to be used in a number of established turnkey projects.<sup>25</sup>

In 2009, First Solar maintained its headquarters in Arizona and operated manufacturing plants in the U.S., Germany and Malaysia. They had 5,200 employees, \$2.1 billion in revenue and profits of \$640 million.<sup>26</sup> At the time First Solar had completed or was under contract to develop solar power generating stations in the U.S. (specifically in the southwest), Germany and the United Arab

Emirates.<sup>27</sup> First Solar's integrated system included project development, construction, asset management, operating and maintenance services, and finance. Project development included the design of solar power-generating plants using First Solar's modules and the procurement of all other components and expertise from third-party suppliers. First Solar also signed long-term and fixed cost agreement with clients for the operations and maintenance of the power plants.<sup>28</sup>

First Solar utilized thin film cadmium telluride in its PV panels. They believed that cost efficiency could be achieved using the cheaper and less efficient material.<sup>29</sup> In 2005, First Solar produced 25 MWs of power with an average cost to build one module of \$1.59. In 2008, production had increased to 716 MW, while the price per module dropped to \$1.08.<sup>30</sup> In the final quarter of 2005, First Solar produced roughly 100,000 PV panels with a conversion efficiency rate of 9.5%. In the first quarter of 2009, First Solar produced almost 300,000 modules with conversion efficiency rates over 10%.<sup>31</sup> Conversely, in 2009, SunPower, the industry leader in c-Si technology, produced modules which converted solar power to electricity with 23% efficiency.<sup>32</sup> First Solar would need almost twice as much physical space to capture the same amount of energy as SunPower.<sup>33</sup>

From its 2006 IPO price of \$20/share First Solar stock reached a high of over \$300/share in mid-2008 before settling to around \$150/share in late March 2009.<sup>34</sup> (See **Exhibit 1** for historical stock prices and **Exhibit 2** for valuation multiples of key competitors.) The initial rise over \$300 was supported by positive coverage from Wall Street analysts who lauded First Solar's performance and market potential. In October 2007, when First Solar was trading at \$135, Morgan Stanley rated the stock as "overweight" and established a 52-week price target of \$175.<sup>35</sup> In June 2008, when First Solar's stock was trading at \$257, Deutsche Bank maintained a price target of \$300 and a recommendation of "buy."<sup>36</sup> Two months later, in response to Q2 earnings, they raised its price target to \$330.<sup>37</sup> Similarly, after the Q2 results, Credit Suisse set a price target at \$330 and rated First Solar as "outperform."<sup>38</sup> The bullish reports focused on First Solar's use of cadmium telluride, which decreased any meaningful competition and increased gross margins at a time of high polysilicon prices, and credited management's ability to leverage First Solar's technological advantages to sign long-term sales contracts.<sup>39</sup> Analysts were also impressed with management's emphasis on production, including expanding facilities to Asia and focusing on throughputs in existing factories. Finally, the reports cited the positive environment of government subsidies that existed in the U.S., Japan and Germany for solar development and investment.<sup>40</sup>

Despite its overall positive rating, analysts also identified areas of potential risk. First, on the technology side they warned that the cadmium telluride technology was new and had not undergone the real-world testing its silicon counterpart had which could lead to future product warranty, performance and safety issues.<sup>41</sup> Second, the reports raised the prospect of uncertainties hampering the successful roll out of the Malaysia plant while keeping operations running smoothly in Germany and Ohio.<sup>42</sup> Third, some analysts speculated that declining polysilicon prices could allow competitors using crystalline silicon technology to reduce the pricing gap, an important First Solar competitive advantage.<sup>43</sup> Fourth, the reports reminded investors that while governments continued to support renewable energies, public policy shifts were not uncommon and could drastically alter First Solar's growth potential.<sup>44</sup> Fifth, analysts were aware of the rising BoS costs associated with building large-scale solar systems and the dwindling opportunities for financing.<sup>45</sup> And finally, some worried that First Solar's international exposure, particularly its growing presence in Germany, created risk from the depreciation of the Euro.<sup>46</sup>

## CFRA Industry Report: Solar Energy<sup>47</sup>

On January 11, 2008 CFRA issued its inaugural solar industry report amidst frequent IPOs and share prices of the dominant firms skyrocketing. The report examined 22 leading solar energy firms. Despite the growth of the industry CFRA identified a number of areas which posed risks to the industry's continued growth and potential financial disappointments for investors. The report was followed up with another industry report in July 2008.

**Revenue quality and recognition** CFRA's prime concern with solar companies was their rapid growth buttressed by long-term customer contracts. At issue was the potential of inconsistent and confusing revenue quality and recognition. The report stated:

RiskMetrics [CFRA used the parent company RiskMetrics name in the reports] generally considers significant increases in trade receivables and/or in unbilled receivables, as measured in days sales, to be indicative of either more generous credit terms or sales incentives or, in certain instances, a more aggressive revenue recognition practice. Similarly, we generally view a decline in deferred revenue, as expressed in day's sales, to be indicative of either a decline in demand for a company's product/services or of an accelerated revenue recognition pace. However, given the rapid revenue growth being experienced by the companies analyzed, we did our best to separate changes in these balance sheet accounts due to growth in the business with those that appear atypical regardless of growth (and are therefore concerning).

The report went on to address the issue of deferred revenue at integrated solar firms, "where companies had a 'project' or 'system' segment which served customers by providing solar equipment and by installing or integrating the equipment at the customer's facilities. The report added that deferred revenues were typical across the solar industry as customers typically paid in advance to secure the product which CFRA considered a sign of healthy demand in the industry."

In these cases, there will also be an 'unbilled receivable' account and a project related deferred revenue account that typically represents billing to customers for which revenues have not yet been recognized. In these cases, we perform another layer of analysis where disclosure is adequate by looking at trends in unbilled receivables and contract deferred revenue. Unfortunately, and of some concern, there is limited disclosure of unbilled receivables among companies that have disclosed the use of the percentage-of-completion (POC) method for revenue recognition on projects

Based on the review, CFRA found that, "Several companies in the solar industry are increasingly reliant on more generous credit terms or other types of financing to drive revenues. RiskMetrics believes that this is noteworthy as it is indicative of the increased competitiveness in the solar market, and also raises concern as to whether the next step will be to reduce pricing to drive sales, which could impair profit margins going forward." (See **Exhibits 3a & 3b** for select company DSO and deferred revenue trends.)

**Inventory levels** While acknowledging that the solar industry was more volatile than others, and that some increase in inventories were expected, CFRA considered the growth of the solar industry and subsequent growth of inventories as another red flag. The report stated:

Risk Metrics considers inventory trends to be a reliable indicator of a company's margin sustainability prospects as well as an indicator of demand for a company's products. We monitor inventory trends by analyzing inventories in day's sales (DSI). Increases in DSI often indicate that demand was weaker than anticipated causing an increase in inventory or that a

company has produced and excess amount of inventory to take advantage of higher fixed costs absorption. In either case, if the company is unable to reduce the excess inventories, margins in future periods will be pressured as production is slowed to pare inventory levels...Similarly, discrepancies between inventory growth trends and expected forward sales growth can also be indicative of weaker than anticipated growth or excess production.

CFRA feared that aggressive inventory growth could put pressure on future margins. (See **Exhibits 4a & 4b** for select company DSI and inventory trends.)

**Limited customer and geographic distribution** CFRA saw risk in solar companies that were too concentrated amongst large customers and/or geographic locations. The report stated:

When a significant portion of a company's revenues are derived from a limited number of customers, RiskMetrics believes that this concentration creates revenue risk. This risk stems from the material loss in business that would occur should a significant customer choose to cease doing business with the company. RiskMetrics believes that customer concentration is a significant risk to several companies in the solar industry. On average, these companies had approximately 3 customers accounting for more than 10% of sales each, and these top 3 customers on average combined to account for approximately 42% of sales. Given the increased competition in the solar industry as companies add capacity and new firms enter the market, the risk of loss of an individual customer is increasing. Buyers of solar products have an increasing number of companies and products to choose from and therefore have an increasing amount of bargaining power with solar providers. Thus, we believe there is significant risk in cases where firms do not have an adequately diversified customer base.

Additionally, many companies in the solar industry have a high exposure to one particular country or geography. The geographic concentration appears to be driven by those countries providing subsidies, feed-in tariffs, or other support programs for solar producers. Specifically, companies with high geographic concentration typically have high exposure to one or more of the following: Germany, Spain, the United States and China. This is a concern to the extent that a country in which a particular company derives a significant portion of its revenues reduces its level of support.

(See **Exhibits 5a** and **5b** for customer concentration of select firms, 2007 and 2008. See **Exhibits 6a** and **6b** for geographic concentrations of select firms, 2007 and 2008.)

**Warranty costs** Most solar companies offered warranties on malfunction and defect of PV panels. In addition, companies guaranteed the long-term power generation of their cells. According to CFRA, "on average the companies surveyed accrued a little over 1% of annual sales for future warranty costs." The report cautioned that the level of warranty provided by many solar companies over periods extending to 25 years created a potential risk to future margins if a quality issue arose. (See **Exhibit 7** for warranty policies of select firms.)

**Capacity growth** In-line with its concern about the sustainability of the growth potential of the solar industry, CFRA identified increasing production capacity as a concern. CFRA was concerned that excess capacity would lead to lower utilization levels and/or lower prices. The report stated:

Across the solar industry, companies are aggressively adding capacity for the future given the expectations for continued growth in demand for solar energy. RiskMetrics raises concern that in certain cases, the growth in capacity may be excessive and margins may deteriorate should demand not keep pace with capacity growth... On average, the companies in this

analysis are planning on increasing capacity by approximately 107% by the end of 2008. While less data is available, companies are on average increasing capacity by an additional 97% in 2009...In total, these firms have targeted total capacity of 6,123 MW by the end of 2008, which appears excessive when one considers that this analysis excludes several companies that have traditionally been the largest solar cell manufacturers. These companies include Sharp, Kyocera, and Sanyo, which all had over 5% of the solar market in 2006, according to a study done by the European Photovoltaic Industry Association ("EPIA")...With regard to total industry capacity requirements; several firms site Solarbuzz, an international solar energy research and consulting company. Annual PV installations industry-wide totaled 1,744 MW in 2006 and are expected to grow to 4,711 MW by 2011 based on Solarbuzz's "Balanced Energy" scenario. Based on this forecast, we feel that there is a great risk that several of these companies may be too aggressively adding capacity, which creates concern that utilization will decline and negatively impact profit margins.

CFRA considered two outcomes, "(1) the excess capacity is never fully put to work resulting in low utilization levels (2) the excess capacity produces greater than needed supply of PV products, resulting in a decline in pricing." (See **Exhibit 8** for production capacity targets.)

**Supply chain** In an increasingly growing and competitive market, CFRA concluded that access to raw material could pose a future threat to expanding companies. The report stated:

Across the solar industry, a shortage of silicon feedstock has caused inventory and order disruption at certain solar energy equipment manufacturing companies, in recent quarters. As most solar cell and PV module manufacturers do not have poly-silicon production capability, companies have proactively managed their future silicon supply needs by entering into long-term supply agreements with silicon feedstock, ingot, wafer and in some cases silicon cell providers. While we believe that the absence of long term supply agreement could result in a company's failure to provide adequate product to customers, we also note that a company locked into a long-term supply contract may be at a disadvantage should spot prices decline to levels below those provided for in the long-term contract.<sup>48</sup>

(See **Exhibit 9** for the description of supply chain integration at select firms.)

## The Solar Industry in Early 2009

While the solar industry was booming in the first half of 2008, a series of events halted its growth by early 2009. Most dramatically, the onset of the financial crisis dampened investor confidence and optimism. As investors lost their appetites for sectors with high perceived risk, financing options for solar customers disappeared. The loss of financing was particularly damaging as high BoS costs required significant upfront financing. In addition, the growing financial crisis also halted the growth of solar in Europe, a key market for the solar industry. Germany and Spain, two of the biggest markets, were forced to cut subsidies to utilities and halt the construction of new projects.<sup>49</sup> Additionally, utilities and policy makers were faced with falling oil and natural gas prices that further reduced the attractiveness of solar as an energy source.

Aside from the financial landscape, the growth of production leading up to 2009 led to dramatic decreases in product prices. An oversupply of raw material coming on line also threatened growth. At the end of 2008 raw material extraction and production, namely of polysilicon, began to gain speed and the once expensive commodity drastically reduced in price. At its peak in 2008, polysilicon commanded anywhere between \$400 and \$500 per kilogram. By February 2009, increased production

had brought prices down to roughly \$150.<sup>50</sup> (See **Exhibit 10** for historical and projected polysilicon prices.)

Despite the challenges, some opportunities also arose. In February, President Obama signed the American Recovery and Reinvestment Act of 2009, more commonly known as the “stimulus bill.” The legislation provided \$44 billion of federal funds for energy-related purposes, provided tax benefits to alternative energy companies, created a \$6 billion loan guarantee program for renewable energy firms and provided \$12 billion in funding for energy efficiency efforts.<sup>51</sup>

Summing up the state of the solar industry at the beginning of 2009, a UBS sector report claimed,

In terms of revenue dollars, we expect average module price declines of 20% and system price (module plus balance of systems) declines of around 15%. We assume there will be an inventory build-up during the year, part of which will count toward module sales. If added up all together, we expect solar revenues will contract 20% from USD 44 billion in 2008 to USD 35 billion in 2009. We expect the largest large portion of this decline to be due to less favorable credit terms, which reduces the economics of solar over the near term. The unexpectedly strong demand from Spain during 2008 and the consequent weak demand in 2009 is also a key reason. On a five-year horizon, we expect the solar industry to grow revenues by a 17% CAGR. If we exclude 2009 (when we expect demand to contract), we forecast the solar industry to post a 28% CAGR in 2010-2013. As a result, we forecast the industry to grow from USD 44 billion in 2008 to about USD 95 billion by 2013.<sup>52</sup>

## First Solar Q1 Results

On April 29<sup>th</sup> 2009, with its stock price just over \$ 151, First Solar released its financials for the first quarter of 2009 and reported revenues of \$418.2 million, beating the consensus estimate by \$18.5 million.<sup>53</sup> Immediately following the publication, analyst reports began heralding First Solar's potential to make up ground despite the post-crisis market sentiment. Analysts' consensus estimate for revenue growth in 2009 was 54.2%.<sup>54</sup> Credit Suisse raised its target price to \$175 and gave a rating of “outperform.” Their evaluation was based on First Solar's higher volumes, continued low production costs and competitive pricing.<sup>55</sup> Specifically, Credit Suisse identified First Solar as a “best-of-breed” company and that its “outperformance is purely a company specific alpha...due to its differentiated, low cost thin technology.”<sup>56</sup> Similarly, Deutsche Bank, who maintained a “hold” rating and a \$167 price target, saw First Solar as best positioned within the industry to take advantage of future opportunities, liked their fundamentals—including low manufacturing costs—and were reassured by their high levels of capacity—two-thirds—under long-term contract.<sup>57</sup> Finally, Deutsche Bank was buoyed by company claims that lending for small projects was increasing in Germany.<sup>58</sup>

Most bullish, and similarly impressed with First Solar's cost reductions and ability to compete on price with industry rivals, were Jefferies & Company analysts who gave First Solar a “buy” rating and a price target of \$215.<sup>59</sup> Jefferies analysts believed that First Solar's ability to continue cutting costs would make it difficult for competitors to catch them on cost, even with the decline of polysilicon prices.<sup>60</sup> Jefferies was also impressed by First Solar's construction efficiency, positing that First Solar's ability to quickly develop and construct projects minimized risk and led to healthy project completion rates.<sup>61</sup>

CFRA was skeptical of such aggressive growth projections, especially following the crisis. CFRA analysts were eager to see how the leadership at First Solar was reporting on, and responding to, the new market realities. They especially wanted to compare First Solar against four companies

considered to be its closest competitors - Evergreen Solar Inc. (ESLR), Q-Cells AG (QCE), SunPower Corp. (SPWR) and Suntech Power Holdings Co. Ltd (STP).

The CFRA team reviewed a handful of previous disclosures from First Solar and highlighted some points of interest and areas of further review. First, the analysts wanted to understand First Solar's current level of receivables. Statements in the December 2008 10-K claimed, "We do not offer extended payment terms or rights of return of sold products."

Second, the same 10-K, addressed long-term contracts between First Solar and its customers. The filing stated that First Solar amended contracts "with two customers to accelerate the decline in the sales price per watt under such contracts in 2009 and 2010 in exchange for increases in the volume of solar modules to be delivered under such contracts."<sup>62</sup> The filing went on to say that First Solar "was in discussion with several other customers about making similar amendments to their Long-Term Supply contracts."<sup>63</sup> On the topic of contracts, the company had previously specified that the contracts required "a sales price per watt that declines by approximately 6.5% at the beginning of each year through the expiration date of each contract in 2012."<sup>64</sup> The CFRA team wanted to understand how the company had changed the contracts and was looking to disclosures in the latest 10-Q. (See **Exhibit 11** for selected passages from First Solar's March 2009 10Q.)

Third, with regard to potential contract defaults, CFRA noted that in the 2008 4<sup>th</sup> quarter earnings conference call First Solar discussed mitigating default risk by "monitoring and trying to create excess demand or options to reallocate those volumes should that become necessary."<sup>65</sup> The analysts were interested in finding out what the company felt was the expected default rate and how the potential for client default was incorporated into the company's financials and guidance for the current year. (See **Exhibit 12** for selected passages from First Solar's Q1 2009 earnings call.)

Fourth, customer concentration was an area of concern. First Solar reported in its 2008 10 K that its principal customers were Blitzstrom GmbH, Colexon Energy AG (previously Reinecke + Pohl), Conergy AG, Juwi Solar GmbH and Phoenix Solar AG. During 2008, each of these five customers individually had accounted for between 11% and 19% of their net sales. (See **Exhibit 13** for First Solar's 2006-2008 customer concentration). The analysts were also able to collect information on inventory and cost of sales data for some of these top five customers (See **Exhibits 14a, 14b and 14c** for select 2007-2009 inventory levels at First Solar's customers.) Of additional interest was an announcement that First Solar had begun co-investing in German solar projects "to alleviate project financing uncertainties" through a joint venture with one of their customers Juwi Solar.<sup>66</sup>

Finally, the analysts studied the company's quarterly income statements, balance sheets and cash flows for the past 11 quarters. (See **Exhibit 15** for First Solar's Income Statement, **Exhibit 16** for First Solar's Balance Sheet, **Exhibit 17** for First Solar's Cash Flow.) While they would have preferred more information and more time, CFRA's analysts, like most forensic accountants, were experts at discerning important trends from limited information. Knowing clients were eagerly awaiting conclusions; the team set out to scrutinize First Solar's revenue quality, inventory levels, capacity issues, customer concentration, geographic distribution and warranty policies.

**Exhibit 1** First Solar's Historical Stock Price (daily, November 16, 2006 to May 1, 2009)

Source: NASDAQ, FSLR Historical Stock Prices, accessed November 9, 2012.

**Exhibit 2** Price-to-Earnings and Price-to-Book Ratios for Selected Solar Companies

|        | P/E    |        |       | P/B   |       |      |
|--------|--------|--------|-------|-------|-------|------|
|        | 2006   | 2007   | 2008  | 2006  | 2007  | 2008 |
| CGY    | 58.96  | 33.23  | 18.98 | 10.75 | 2.59  | 2.11 |
| CSIQ   | -      | -      | -     | 8.37  | 6.63  | 0.59 |
| CSUN   | 30.67  | 30.59  | -     | 12.95 | 12.92 | 0.79 |
| ESLR   |        |        |       | 5.64  | 4.52  | 0.90 |
| FSLR   | 241.61 | 194.74 | 40.51 | 13.19 | 21.23 | 8.08 |
| STP    | 60.16  | 88.39  | 8.79  | 8.32  | 15.37 | 1.61 |
| SWV.DE | 21.91  | 44     | 11.76 | 4.38  | 7.05  | 2.08 |
| TSL    | 20.87  | 47.76  | 2.96  | 9.27  | 3.88  | 0.54 |
| YGE    | 167.62 | 198.45 | 7.6   | 33.94 | 9.56  | 1.15 |

Source: Capital IQ, accessed December 4, 2012.

Note: All dates as of December 31<sup>st</sup>. P/E ratios are unavailable for companies incurring a loss.

**Exhibit 3a** Days Sales Outstanding at Selected Solar Firms

|      | Mar-09 | Dec-08 | Sep-08 | Jun-08 | Mar-08 | Dec-07 | Sep-07 | Jun-07 | Mar-07 | Dec-06 | Sep-06 | Jun-06 | Mar-06 |
|------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| CGY  | 95     | 54     | 51     | 57     | 67     | 33     | 194    | 150    | 229    | 77     | 163    | 127    | 146    |
| CSIQ | 125    | 65     | 55     | 60     | 64     | 42     | 46     | 59     | 105    | 65     | 18     | 32     | -      |
| CSUN | 23     | 19     | 12     | 5      | 11     | 34     | 48     | 62     | 46     | -      | -      | -      | -      |
| ESLR | 78     | 73     | 50     | 37     | 52     | 38     | 55     | 68     | 89     | 71     | 57     | 76     | 46     |
| QCE  | 111    | 52     | 44     | 40     | 51     | 39     | 45     | 24     | 48     | 48     | 42     | 30     | 31     |
| SOLF | 27     | 26     | 25     | 30     | 51     | 40     | 83     | 71     | 114    | 55     | 3      | -      | -      |
| SPWR | 64     | 44     | 47     | 59     | 53     | 56     | 32     | 49     | 53     | 58     | 65     | 57     | 70     |
| STP  | 62     | 47     | 36     | 42     | 57     | 55     | 56     | 65     | 61     | 41     | 11     | 21     | 5      |
| YGE  | 173    | 76     | 48     | 67     | 67     | 78     | 93     | 59     | 54     | 47     | 9      | -      | -      |

**Exhibit 3b** Deferred Revenue Days Trends at Selected Solar Firms

|      | Mar-09 | Dec-08 | Sep-08 | Jun-08 | Mar-08 | Dec-07 | Sep-07 | Jun-07 | Mar-07 | Dec-06 | Sep-06 | Jun-06 | Mar-06 |
|------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| CSIQ | 17     | 5      | 3      | 5      | 4      | 1      | 9      | 12     | 4      | 12     | 13     | 39     | -      |
| CSUN | -      | 1      | 1      | 6      | 7      | 6      | 4      | 7      | 9      | -      | -      | -      | -      |
| SOLF | 0      | 1      | 2      | 5      | 7      | 2      | 10     | 0      | 0      | 0      | 8      | -      | -      |
| SPIR | -      | -      | -      | -      | -      | -      | 224    | 148    | 120    | 148    | 97     | 37     | 18     |
| SPWR | 14     | 8      | 5      | 11     | 14     | 51     | 12     | 43     | 45     | 32     | 75     | -      | -      |
| YGE  | 5      | 2      | 1      | 0      | 4      | 3      | 1      | 4      | 14     | 19     | -      | -      | -      |

Source for 3a and 3b: Timothy Rickert, "Diversified Industrials—Solar Energy," CFRA Industry Report, 2009, Center for Financial Research and Analysis, ISS Inc.

Note: (CGY) Conergy AG; (CSIQ) Canadian Solar Inc.; (CSUN) China Sunenergy Co Ltd. ADS; (ESLR) Evergreen Solar Inc.; (FSLR) First Solar Inc.; (QCE) Q-Cells AG; (SOLF) Solarfun Power Holdings Co. Ltd.; (SPWR) SunPower Corp.; (STP) Sutech Power Holdings Co. Ltd.; (YGE) Yingli Green Energy Holding Co. Ltd.

**Exhibit 4a** Days Sales Inventory at Selected Solar Firms

|      | Mar-09 | Dec-08 | Sep-08 | Jun-08 | Mar-08 | Dec-07 | Sep-07 | Jun-07 | Mar-07 | Dec-06 | Sep-06 | Jun-06 | Mar-06 |
|------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| CGY  | 484    | 121    | 117    | 71     | 220    | 64     | 157    | 92     | 165    | 42     | 152    | 73     | 87     |
| CSIQ | 173    | 86     | 43     | 46     | 52     | 57     | 66     | 94     | 259    | 149    | 211    | 196    | -      |
| CSUN | 59     | 94     | 35     | 38     | 70     | 77     | 111    | 81     | 75     | 72     | -      | -      | -      |
| ESLR | 49     | 51     | 54     | 47     | 31     | 46     | 50     | 38     | 44     | 17     | 37     | 51     | 43     |
| QCE  | 123    | 75     | 75     | 69     | 43     | 41     | 55     | 46     | 86     | 76     | 68     | 72     | 78     |
| SOLF | 108    | 61     | 56     | 64     | 70     | 95     | 75     | 96     | 279    | 190    | 76     | -      | -      |
| SPWR | 189    | 79     | 64     | 63     | 78     | 72     | 47     | 64     | 61     | 37     | 47     | 46     | 44     |
| STP  | 82     | 51     | 49     | 46     | 48     | 51     | 49     | 72     | 87     | 108    | 44     | 73     | 90     |
| YGE  | 254    | 122    | 102    | -      | 108    | 105    | 99     | 125    | 242    | 177    | -      | -      | -      |

Source: Timothy Rickert, "Diversified Industrials – Solar Energy," CFRA Industry Report, September 14, 2009. Center for Financial Research and Analysis, ISS Inc.

**Exhibit 4b** Inventory Growth (January 2008)

|                       | <b>AKNS</b> | <b>CGY</b>  | <b>CSIQ</b> | <b>CSUN</b> | <b>ESLR</b> | <b>FSLR</b> |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Inventory Growth, YOY | 345%        | 81%         | 126%        | 32%         | -40%        | 203%        |
| NTM Sales Growth      | 116%        | 29%         | 244%        | 74%         | 29%         | 124%        |
| Inventory/NTM Growth  | 230%        | 52%         | -117%       | -42%        | -69%        | 79%         |
|                       | <b>QCE</b>  | <b>SOLF</b> | <b>SPIR</b> | <b>SPWR</b> | <b>STP</b>  | <b>YGE</b>  |
| Inventory Growth, YOY | 37%         | 249%        | 452%        | 283%        | 171%        | 61%         |
| NTM Sales Growth      | 117%        | 172%        | -           | 95%         | 89%         | 127%        |
| Inventory/NTM Growth  | -30%        | 77%         | -           | 188%        | 82%         | -67%        |

Source: Dan Mahoney, Jean-Louis Thiémélé, Jon Dye, "Diversified Industrials - Solar Energy," CFRA Industry Report, January 11, 2008. Center for Financial Research and Analysis, ISS Inc.

Note: (YOY) Year-on-year; (NTM) next-twelve-months. (CGY) Conergy AG; (CSIQ) Canadian Solar Inc.; (CSUN) China Sunenergy Co Ltd. ADS; (ESLR) Evergreen Solar Inc.; (FSLR) First Solar Inc.; (QCE) Q-Cells AG; (SOLF) Solarfun Power Holdings Co. Ltd.; (SPWR) SunPower Corp.; (STP) Sutech Power Holdings Co. Ltd.; (YGE) Yingli Green Energy Holding Co. Ltd.

**Exhibit 5a** Customer Concentration, Year End December 2007

|                                           | <b>AKNS</b> | <b>CGY</b>  | <b>CSIQ</b> | <b>CSUN</b> | <b>ESLR</b> | <b>FSLR</b> |
|-------------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| % of Sales to Largest Customer            | -           | -           | 21.1%       | 24%         | 31%         | 22.6%       |
| # of Customers w/ > 10% Sales             | -           | -           | 4           | 2           | 3           | 6           |
| % of Sales to Customers w/ > 10% of Sales | -           | -           | 75.3%       | 41.3%       | 57%         | 88.5%       |
|                                           | <b>QCE</b>  | <b>SOLF</b> | <b>SPIR</b> | <b>SPWR</b> | <b>STP</b>  | <b>YGE</b>  |
| % of Sales to Largest Customer            | -           | 12.6%       | -           | 18%         | 20.2%       | 20%         |
| # of Customers w/ > 10% Sales             | -           | 1           | -           | 2           | 3           | 3           |
| % of Sales to Customers w/ > 10% of Sales | -           | 12.6%       | -           | 34%         | 47.9%       | 45%         |

Source: Dan Mahoney, "Diversified Industrials - Solar Energy," CFRA Industry Report, July 22, 2008. Center for Financial Research and Analysis, ISS Inc.

**Exhibit 5b** Customer Concentration, Year End December 2008

|                                           | <b>AKNS</b> | <b>CGY</b>  | <b>CSIQ</b> | <b>CSUN</b> | <b>ESLR</b> | <b>FSLR</b> |
|-------------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| % of Sales to Largest Customer            | -           | -           | 14.7%       | 18%         | 52%         | 18.6%       |
| # of Customers w/ > 10% Sales             | -           | -           | 3           | 3           | 2           | 5           |
| % of Sales to Customers w/ > 10% of Sales | -           | -           | 38.1%       | 46%         | 76%         | 64.1%       |
|                                           | <b>QCE</b>  | <b>SOLF</b> | <b>SPIR</b> | <b>SPWR</b> | <b>STP</b>  | <b>YGE</b>  |
| % of Sales to Largest Customer            | -           | 22.2%       | -           | 22%         | 10%         | 11.6%       |
| # of Customers w/ > 10% Sales             | -           | 1           | -           | 1           | -           | 1           |
| % of Sales to Customers w/ > 10% of Sales | -           | 22.2%       | -           | 22%         | -           | 11.6%       |

Source: Timothy Rickert, "Diversified Industrials – Solar Energy," CFRA Industry Report, September 14, 2009. Center for Financial Research and Analysis, ISS Inc.

**Exhibit 6a** Geographic Concentration, FY 2007

|                    | <b>AKNS</b> | <b>CGY</b>  | <b>CSIQ</b> | <b>CSUN</b> | <b>ESLR</b> | <b>FSLR</b> |
|--------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| U.S./North America | -           | -           | -           | -           | 82%         | -           |
| Germany            | -           | 48%         | 68%         | 27%         | 7%          | 91%         |
| Spain/Europe       | -           | 33%         | 26%         | 1%          | -           | -           |
| China/Asia         | -           | -           | 6%          | 67%         | -           | -           |
| Rest of World      | -           | 19%         | -           | 5%          | 11%         | 9%          |
|                    | <b>QCE</b>  | <b>SOLF</b> | <b>SPIR</b> | <b>SPWR</b> | <b>STP</b>  | <b>YGE</b>  |
| U.S./North America | 6%          | 1%          | -           | 45%         | 6%          | -           |
| Germany            | 39%         | 50%         | -           | 10%         | 51%         | 22%         |
| Spain/Europe       | 39%         | 28%         | -           | 40%         | 38%         | 72%         |
| China/Asia         | 9%          | 8%          | -           | -           | 3%          | 6%          |
| Rest of World      | 7%          | 13%         | -           | 5%          | 2%          | -           |

**Exhibit 6b** Geographic Concentration, FY 2008

|                    | <b>AKNS</b> | <b>CGY</b>  | <b>CSIQ</b> | <b>CSUN</b> | <b>ESLR</b> | <b>FSLR</b> |
|--------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| U.S./North America | -           | -           | 5%          | -           | 53%         | -           |
| Germany            | -           | 29%         | 62%         | 36%         | 10%         | 74%         |
| Spain/Europe       | -           | 50%         | 27%         | 5%          | 27%         | 20%         |
| China/Asia         | -           | -           | 6%          | 59%         | 3%          | -           |
| Rest of World      | -           | 21%         | -           | -           | 7%          | 6%          |
|                    | <b>QCE</b>  | <b>SOLF</b> | <b>SPIR</b> | <b>SPWR</b> | <b>STP</b>  | <b>YGE</b>  |
| U.S./North America | -           | -           | -           | 55%         | 7%          | -           |
| Germany            | -           | 53%         | -           | 17%         | 30%         | 41%         |
| Spain/Europe       | -           | 40%         | -           | 22%         | 48%         | 47%         |
| China/Asia         | -           | 7%          | -           | -           | 7%          | 10%         |
| Rest of World      | -           | -           | -           | 6%          | 8%          | 2%          |

Source for 6a and 6b: Timothy Rickert, "Diversified Industrials – Solar Energy," CFRA Industry Report, September 14, 2009. Center for Financial Research and Analysis, ISS Inc.

**Exhibit 7** Warranty Policy of Selected Solar Firms, 2008

|      | <b>Warranty Policy</b>                                                                                                                                                                            | <b>Liability/<br/>TTM<br/>Sales</b> |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| AKNS | Product defect / workmanship warranty: 5 years<br>Power generation warranty: 5 to 25 years on solar panel and inverters<br>Accrual policy: Based on historical experience and future expectations | 2.2%                                |
| CGY  | -                                                                                                                                                                                                 | 1.98%                               |
| CSIQ | Product defect/workmanship: 2 years<br>Power generation warranty: 90% for 10 years - 80% for 25 years<br>Accrual policy: 1% of sales                                                              | 1.54%                               |
| CSUN | Product defect / workmanship warranty: Not disclosed<br>Power generation warranty: 10 or 20 years on certain sales contracts<br>Accrual policy: 0.5% of sales                                     | -                                   |
| ESLR | Product defect / workmanship warranty: 2 years<br>Power generation warranty: 25 years<br>Accrual policy: Used historical failure rates                                                            | 1.07%                               |
| FSLR | Product defect / workmanship warranty: 5 years<br>Power generation warranty: 90% for 10 years - 80% for 25 years<br>Accrual policy: Not disclosed                                                 | 0.96%                               |
| QCE  | Product defect / workmanship warranty: Not disclosed<br>Power generation warranty: 90% for 10 years - 80% for 20 years<br>Accrual policy: Not disclosed                                           | 0.66%                               |
| SOLF | Product defect / workmanship warranty: 2 years<br>Power generation warranty: 90% for 10 years - 80% for 25 years<br>Accrual policy: 1% of net revenue                                             | 0.96%                               |
| SPIR | Product defect / workmanship warranty: 1 year<br>Power generation warranty: 80% for 20 years<br>Accrual policy: Not disclosed                                                                     | 0.6%                                |
| SPWR | Product defect / workmanship warranty: At least 1 year<br>Power generation warranty: Up to 25 years<br>Accrual policy: Accrual for known and for incurred but not reported product issues         | 1.96%                               |
| STP  | Product defect / workmanship warranty: Not disclosed<br>Power generation warranty: Up to 25 years<br>Accrual policy: Accrued based on competitor's history and internal failure estimates.        | 2.2%                                |
| YGE  | Product defect / workmanship warranty: 2 years<br>Power generation warranty: 90% for 10 years - 80% for 25 years<br>Accrual policy: 1% of revenues                                                | 1.64%                               |

Source: Dan Mahoney, Jean-Louis Thiémélé, Jon Dye, "Diversified Industrials—Solar Energy," CFRA Industry Report, January 11, 2008 & Timothy Rickert, "Diversified Industrials—Solar Energy," CFRA Industry Report, September 14, 2009. Center for Financial Research and Analysis, ISS Inc.

**Exhibit 8** Production Capacity Targets at Selected Solar Firms (megawatts)

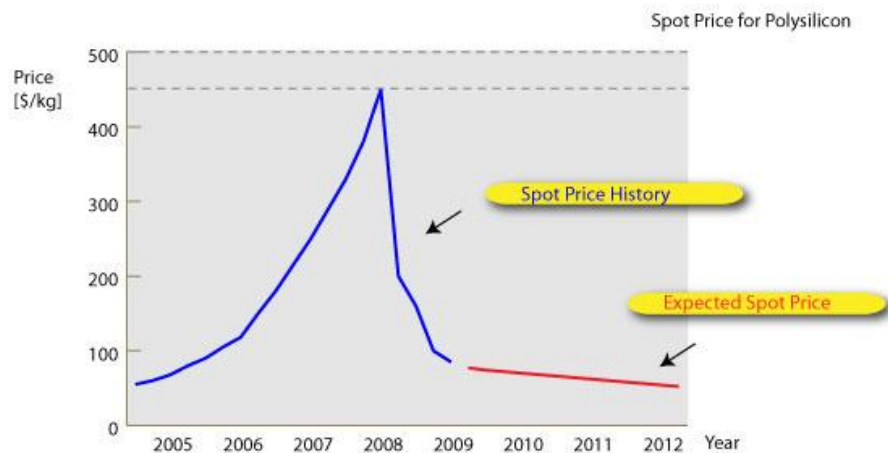
|      | 2007 | 2008  |
|------|------|-------|
| AKNS | -    | -     |
| CGY  | -    | -     |
| CSIQ | 100  | 250   |
| CSUN | 192  | 320   |
| ESLR | 15   |       |
| FSLR | 277  | 435   |
| QCE  | 516  | -     |
| SOLF | 240  | 360   |
| SPIR | -    | -     |
| SPWR | 174  | 414   |
| STP  | 540  | 1,000 |
| YGE  | 200  | 400   |

Source: Dan Mahoney, "Diversified Industrials – Solar Energy," CFRA Industry Report, July 22, 2008. Center for Financial Research and Analysis, ISS Inc.

**Exhibit 9** Supply Chain Integration at Solar Firms

|      | Solar<br>Cell/Module<br>Manufacturing<br>Equipment | Silicon<br>Feedstock<br>Refining | Raw<br>Material<br>Conversion | Solar Cell<br>Development | PV Panel &<br>Module<br>Development | Turnkey<br>System<br>Development | Turnkey<br>System<br>Installation |
|------|----------------------------------------------------|----------------------------------|-------------------------------|---------------------------|-------------------------------------|----------------------------------|-----------------------------------|
| CGY  |                                                    |                                  |                               | x                         | x                                   | x                                | x                                 |
| CSIQ |                                                    | x                                |                               | x                         | x                                   |                                  |                                   |
| CSUN |                                                    |                                  |                               | x                         |                                     |                                  |                                   |
| ESLR |                                                    |                                  | x                             | x                         | x                                   |                                  |                                   |
| FSLR |                                                    |                                  |                               |                           | x                                   |                                  |                                   |
| QCE  |                                                    |                                  |                               | x                         |                                     |                                  |                                   |
| SOLF |                                                    |                                  | x                             | x                         | x                                   |                                  |                                   |
| SPWR |                                                    |                                  |                               | x                         | x                                   | x                                | x                                 |
| STP  |                                                    |                                  |                               | x                         | x                                   |                                  |                                   |
| YGE  |                                                    |                                  | x                             | x                         | x                                   | x                                | x                                 |

Source: Timothy Rickert, "Diversified Industrials – Solar Energy," CFRA Industry Report, September 14, 2009. Center for Financial Research and Analysis, ISS Inc.

**Exhibit 10** Polysilicon Prices (\$/kg)

Source: Green Rhino Energy website, "Solar Power: Polysilicon," [http://www.greenrhinoenergy.com/solar/industry/ind\\_01\\_silicon.php](http://www.greenrhinoenergy.com/solar/industry/ind_01_silicon.php), accessed November 2012.

**Exhibit 11** Selected Passages from First Solar's 2009 10-Q (May 1, 2009)

### Market Overview

The solar industry has been moving from a supply driven to a demand driven environment, with increasing competitive pressure as the photovoltaic industry's total manufacturing capacity to produce solar modules exceeds current demand for those products. Our customer's face significant challenges under the current economic conditions, including an increase in interest or lending rates or tightening of the supply of capital to finance solar projects. Our net sales could be adversely impacted if legislation reduces the current subsidy programs in Europe, North America or Asia or if interest rates increase or financing is no longer available, which could impact our end-users' ability to either meet their target return on investment or finance their projects. In addition, subsidies for our customers, particularly in Germany, are declining at a rate that is greater than the annual contractual price decline we extend under our Long Term Supply Contracts. As result, we may be less competitive and not meet our customers internal rate of return (IRR) requirements or the profit margin of our customers might decline, which could lower demand for our solar modules.

We face intense competition from manufacturers of crystalline silicon solar modules as well as other types of solar modules and photovoltaic systems. Solar module manufacturers compete with one another in several product performance attributes, including module cost per-watt and levelized cost of electricity, meaning the net present value of total life cycle costs of the solar power project divided by the quantity of energy produced over the system life. We are one of the lowest cost module manufacturers in the solar industry, as evidenced by the further reduction in our average manufacturing cost per watt from \$0.98 in the three months ended December 27, 2008 to \$0.93 in the three months ended March 28, 2009. This cost advantage is reflected in the price at which we sell our modules or fully integrated systems and enables our systems to compete favorably on the levelized cost of electricity generated by our systems. Our cost competitiveness is based on our proprietary technology, which provides lower cost from a continuous highly automated industrial manufacturing process, our scale and our operational excellence. In addition, our modules use approximately 1% of the amount of semiconductor material that is used to manufacture traditional crystalline silicon solar modules. The cost of polysilicon is a significant driver of the manufacturing cost of crystalline silicon solar modules. The current spot market price of polysilicon of approximately \$80 and \$100 per kilogram (Kg) is at a level that enables us to remain one of the lowest cost module manufacturers in the solar industry. However, the timing and velocity of decrease in

the cost of silicon feedstock could lead to pricing pressure for solar modules. Although we are not a crystalline silicon module manufacturer, we estimate based on industry research and public disclosures of our competitors, that a \$10 per Kg increase or decrease in the price of polysilicon could increase or decrease, respectively, our competitors' manufacturing cost per watt by approximately \$0.07 to \$0.08. Given the lower conversion efficiency of our modules compared to crystalline silicon modules, there are higher balance-of-system costs associated with systems using our modules. Thus, to compete effectively on the basis of levelized cost of electricity our modules need to maintain a certain cost advantage per watt compared to crystalline silicon based modules. Our cost reduction roadmap anticipates manufacturing cost per watt reductions for our product of 10% per year. In 2008, we reduced our manufacturing cost per watt by 12%

### **Net Sales**

Currently, the majority of our net sales is generated from the sale of solar modules. We price and sell our solar modules per watt of power. As a result, our net sales can fluctuate based on our output of sellable watts or price. We currently sell almost all of our solar modules to solar power system project developers, system integrators and operators headquartered in Germany, France and Spain, which either resell our solar modules to end-users or integrate them into power plants that they own or operate or sell.

In April 2006, we entered into long-term contracts for the purchase and sale of our solar modules with six European solar power system project developers and system integrators, and in 2007, we entered into additional long-term contracts for the purchase and sale of our solar modules with six other European project developers that also own and operate renewable energy projects. In 2008, we entered into long-term contracts with three European project developers, system integrators and operators and increased our contracted volume with four customers. We also entered into a five-year agreement with a solar power system project developer and system integrator in the United States, which is a related party. These contracts account for a significant portion of our planned production over the period from 2009 through 2013, and therefore, will significantly affect our overall financial performance.

During the three months ended March 28, 2009 we amended our Long Term Supply Contracts for certain customers to accelerate the decline in the sales price per watt under such contracts in 2009 and 2010 in exchange for increases in the volume of solar modules to be delivered under such contracts. We also extended the payment terms for certain customers under these contracts from 10 days to 45 days, net.

### **Gross profit**

Gross profit is affected by numerous factors, including our average selling prices, foreign exchange rates, our manufacturing costs and the effective utilization of our production facilities. For example, our Long Term Supply Contracts specify a sales price per watt that declines 6.5% annually. Other factors impacting gross profits are the ramp of production on new plants due to a reduced ability to absorb fixed costs until full production volumes are reached as well as the mix of net sales generated by our module and project business coupled with a geographic factor. As a result, gross profits may vary from quarter to quarter and year to year.

### **Results of Operations**

Our average selling price decreased by approximately 13% during the three months ended March 28, 2009 compared with the three months ended March 29, 2008, mainly due to an 8% reduction in our average selling price due to annual contractual price declines that took effect in the first quarter of 2009 and amendments made to our Long Term Supply contracts with certain customers in the first quarter of 2009. In addition, our average selling price was adversely impacted by 5% due to an unfavorable decline in the foreign exchange rate between the U.S. dollar and the euro. During the three months ended March 29, 2008, 61.8% of our net sales resulted from sales of solar modules to customers headquartered in Germany.

### **Cash Flows**

Cash provided by operating activities was \$63.7 million during the three months ended March 28, 2009 compared with \$63.3 million during the three months ended March 29, 2008. Accounts receivable increased by \$93.2 million during the three months ended March 28, 2009 mainly due to the amendment of certain customers'

long-term solar supply contracts, that extended our customers' payment terms from 10 days to 45 days, net as well as the timing of shipments to customers during the three months ended March 28, 2009.

### ***Product warranties***

We offer warranties on our products and record an estimate of the associated liability based on the number of solar modules under warranty at customer locations, our historical experience with warranty claims, our monitoring of field installation sites, our in-house testing of our solar modules and our estimated per-module replacement cost.

Product warranty activity during the three months ended March 28, 2009 and March 29, 2008 was as follows (in thousands):

|                                                       | March 28, 2009 | December 27, 2008 |
|-------------------------------------------------------|----------------|-------------------|
| Product warranty liability, beginning of period       | 11,905,        | 7,276             |
| Accruals for new warranties issues (warranty expense) | 3,091          | 1,992             |
| Settlements                                           | (208)          | (8)               |
| Change in estimate of warranty liability              | (1,231)        | 1                 |
| Product warranty liability, end of period             | 13,557         | 9,261             |
| Current portion of warranty                           | 5,311          | 3,921             |
| Non-current portion of warranty liability             | 8,246          | 5,340             |

Source: First Solar Inc., Form 10-Q, published May 5, 2009.

## **Exhibit 12** Selected Passages from First Solar's 2009 Q1 Earning Call

### ***Michael Ahearn, Chief Executive Officer***

Geographically, Germany, Italy, France, Spain and now Ontario, Canada are our PV subsidized markets. Our offering in these markets is a sale of modules as opposed to a more expanded offering. The customers we sell to consider us relative to other PV suppliers, so our competitors are obviously PV module manufacturers and the pricing trend that we would expect over time is one of declining prices in keeping with declining subsidies. It's really PV-specific subsidies that drive demand in these situations. Those subsidies have to come down over time, and we believe that's a political reality. And so do prices therefore.

So, let me turn to some of the risk factors here. If you go to slide eight, one risk, which we've talked about on prior calls, is that we'll see defaults under existing contracts with customers. We continue to see this impacting potentially 10 to 15% of our planned 2009 shipments. Situation really hasn't changed since last quarter. I suppose the passage of time could be marginally helpful, but this scenario we're watching very closely. The mitigation in our case has to do with monitoring and trying to create excess demand or options to reallocate those volumes should that become necessary. So we'd continue on the watch list in that respect.

So, let me go to slide nine, the third big risk here is of course, oversupply in the marketplace and that's been reported and talked about a lot. There are some potential issues here in terms of short-term price reduction in polycrystalline silicon impacting First Solar. And this – if you look to the right column of the slide, you'll see our view on this continues to be that it is not impacting us in the near-term.

Longer-term, the question arises, could polycrystalline silicon cost reduce to the point that your competitive cost advantage is eroded. And that's something we want to be very sensitive to and not have our heads in the sand on. .... Our approach on this is to compare a hypothetical best case on polycrystalline silicon to our own cost reduction roadmap. And we've talked about our cost reduction roadmap previously, but basically it calls for a manufacturing cost in the area of \$0.65 a watt by 2012. There are a number of things we have to execute to get to that point. There are also levers driving this that we think continue beyond 2012 in terms of technology-driven conversion efficiency improvements; scale, driven by very low variable costs, and productivity continues to

drive throughput.....So what this tells us from a, sort of a mitigation point of view is we've got to be focused on executing our cost reduction roadmap. If we do what we've done in the past, and we tend to our knitting, we can continue to move down the path here, open new markets and potentially widen the gap relative to crystalline silicon type offerings.

*Jens Meyerhoff, Chief Financial Officer*

Thank you, Mike, and good afternoon. On slide 15 you see our net sales for the first quarter were 418.2 million, a decrease of 3.6% over the fourth quarter of 2008. The decrease was primarily driven by price declines, customer mix, as well as by sequential decline in the euro exchange rate from a blended \$1.41 per euro in the first quarter to \$1.39 per euro in the first quarter of 2009. These decreases were partially offset by higher shipping volumes from the ramp of plant two and three in Malaysia and higher line throughput.

Our free cash flow on slide 23 was negative by \$22.7 million in the first quarter due to the one-time impact of changing our payment terms from 10 days to 45 days and higher shipment rates in the second half of the first quarter. Cash flow from operations during the first quarter was \$63.7 million driven by continued revenue growth and offset by increased accounts receivable balance and inventories to support growth.

As discussed by Mike, we're monitoring and mitigating certain business risks such as potential customer contract default risk due to either financial instability of customers, limited access to product financing or near-term competitive forces. Such risks are as of today and not incorporated into our guidance due to our belief that our mitigation plans are sufficient. In addition, we may be able to reach the required milestones and trigger events that allow revenue recognition of some of the deferred revenues excluded from our guidance as a possible further risk mitigation path.

Before I get to the guidance for 2009, I would like to provide an update on our activity of committing capital towards financing projects in Europe as shown on slide 25. During the first quarter, we contributed €10.3 million to support the financing of the world's second largest PV plant in Germany. With this commitment we were able, together our customer Juwi, to obtain favorable financing terms for over 80% of the project capital in a challenging project finance market. Due to the structure and timing of the financing, we deferred approximately \$27 million of net sales during the first quarter. The intent, as you see on slide 26, is to sell this attractive project together with Juwi to a long-term equity investor, at which point we plan to represent less than €10 million of the project capital structure.

Now to our guidance on slide 29, we maintain our previous net sales guidance range of 1.9 to \$2 billion. GAAP operating margin is expected between 31 and 33% reflecting additional expenses of the OptiSolar acquisition. We expect our tax rate to be between nine and 11%. Year-end 2009 fully diluted share count guidance is an estimated 86 to 87 million shares.

Source: First Solar Inc. Q1 Earnings Call Transcript.

**Exhibit 13** First Solar's Customer Concentration, 2006–2008

| (\$ Millions, Exc. %) | 2008      |            | 2007      |            | 2006      |            |
|-----------------------|-----------|------------|-----------|------------|-----------|------------|
|                       | Net Sales | % of Total | Net Sales | % of Total | Net Sales | % of Total |
| Customer #1           | 138.8     | 11.1%      | 74.5      | 14.8%      | 23.0      | 17.1%      |
| Customer #2           | 135.2     | 10.9%      | 52.0      | 10.3%      | 21.6      | 16.0%      |
| Customer #3           | 143.9     | 11.5%      | 76.7      | 15.2%      | 25.9      | 19.2%      |
| Customer #4           | 231.6     | 18.6%      | 113.7     | 22.6%      | 25.1      | 18.6%      |
| Customer #5           | 149.9     | 12.0%      | 68.5      | 13.6%      | 22.4      | 16.6%      |
| Customer #6           |           |            | 65.4      | 13.0%      |           |            |
| Total                 |           | 64.1%      |           | 89.5%      |           | 87.5%      |

Source: Dino Sadikovic, "First Solar Inc.," CFRA Company Report, June 10, 2009. Center for Financial Research and Analysis, ISS Inc.

**Exhibit 14a** Conergy's Inventory Levels

| (€ Millions, Exc. DSI) | Q1,<br>Mar-09 | Q2,<br>Mar-08 | Q3,<br>Mar-07 | Q4,<br>Dec-08 | Q4,<br>Dec-07 | Q3,<br>Sep-08 | Q3,<br>Sep-07 | Q2,<br>Jun-08 | Q2,<br>Jun-07 |
|------------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| Inventory              | 242.3         | 402.2         | 203.7         | 245.6         | 345.6         | 248.1         | 284.7         | 251.9         | 261.8         |
| COGS                   | 45.7          | 170.9         | 116.3         | 185.2         | 187.7         | 194.5         | 159.7         | 332.5         | 154.4         |
| DSI                    | 484           | 215           | 160           | 121           | 168           | 116           | 163           | 69            | 155           |

**Exhibit 14b** Phoenix Solar's DSI YoY

| (€ Millions, Exc. DSI) | Q1,<br>Mar-09 | Q2,<br>Mar-08 | Q3,<br>Mar-07 | Q4,<br>Dec-08 | Q4,<br>Dec-07 | Q3,<br>Sep-08 | Q3,<br>Sep-07 | Q2,<br>Jun-08 | Q2,<br>Jun-07 |
|------------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| Inventory              | 108.0         | 55.4          | 36.3          | 76.0          | 25.4          | 87.9          | 53.5          | 68.1          | 43.8          |
| COGS                   | 36.2          | 36.8          | 13.3          | 105.7         | 108.2         | 124.7         | 60.3          | 89.4          | 39.1          |
| DSI                    | 273           | 138           | 248           | 66            | 21            | 64            | 81            | 70            | 102           |

**Exhibit 14c** Colexon's DSI YoY

| (€ Millions, Exc. DSI) | Q1,<br>Mar-09 | Q2,<br>Mar-08 | Q3,<br>Mar-07 | Q4,<br>Dec-08 | Q4,<br>Dec-07 | Q3,<br>Sep-08 | Q3,<br>Sep-07 | Q2,<br>Jun-08 | Q2,<br>Jun-07 |
|------------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| Inventory              | 40.8          | 8.4           | 11.0          | 18.4          | 8.2           | 11.7          | 11.3          | 11.1          | 9.9           |
| COGS                   | 15.7          | 21.7          | 10.9          | 30.6          | 30.0          | 29.9          | 16.3          | 20.8          | 10.3          |
| DSI                    | 238           | 35            | 92            | 55            | 25            | 36            | 63            | 49            | 88            |

Source for 14a, 14b and 14c: Dino Sadikovic, "First Solar Inc.," CFRA Company Report, June 10, 2009. Center for Financial Research and Analysis, ISS Inc.

**Exhibit 15** Income Statement

|                                     | 28-MAR-09 | 27-DEC-08 | 27-SEP-08 | 28-JUN-08 | 29-MAR-08 | 29-DEC-07 | 29-SEP-07 | 30-JUN-07 | 31-MAR-07 | 30-DEC-06 | 30-SEP-06 | 01-JUL-06 | 01-APR-06 |
|-------------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| Total Revenue                       | 418.2     | 433.7     | 348.7     | 267.0     | 196.9     | 200.8     | 159.0     | 77.2      | 66.9      | 52.7      | 40.8      | 27.9      | 13.6      |
| Cost of Sales                       | 182.9     | 199.7     | 153.3     | 122.3     | 92.6      | 89.8      | 77.0      | 48.9      | 36.9      | 27.1      | 24.5      | 18.8      | 10.4      |
| Gross Profit                        | 235.3     | 233.9     | 195.4     | 144.7     | 104.3     | 111.0     | 82.0      | 28.4      | 30.0      | 25.6      | 16.3      | 9.1       | 3.3       |
| Research and development            | 11.7      | 11.1      | 10.0      | 7.7       | 4.8       | 4.4       | 3.9       | 3.8       | 3.1       | 1.6       | 1.7       | 1.5       | 1.5       |
| Selling, general and administrative | 47.8      | 52.7      | 49.0      | 43.6      | 28.7      | 24.2      | 27.1      | 17.3      | 13.7      | 11.0      | 8.4       | 8.1       | 5.9       |
| Acquisition related                 | 1.5       |           |           |           |           |           |           |           |           |           |           |           |           |
| One-time expense                    | 6.2       | 8.8       | 6.3       | 4.6       | 12.8      | 4.1       | 2.8       | 1.5       | 8.5       | 4.0       | 1.1       | 4.1       | 2.6       |
| Production start-up                 | 168.1     | 161.3     | 130.2     | 88.7      | 58.1      | 78.3      | 48.3      | 5.8       | 4.8       | 9.0       | 5.1       | (4.6)     | (6.7)     |
| Operating Income                    |           |           |           |           |           |           |           |           |           |           |           |           |           |
| Foreign currency gain               | 1.8       | 6.2       | (1.9)     | 0.6       | 0.8       | 1.2       | 1.0       | 0.0       | (0.3)     | 2.8       | (0.3)     | 2.2       | 0.9       |
| Interest Income                     | 2.1       | 4.2       | 5.3       | 4.9       | 6.7       | 7.2       | 5.3       | 3.8       | 4.1       |           | 0.3       |           |           |
| Other expense, net                  | (1.3)     | 0.2       | (0.4)     | (0.4)     | (0.4)     | (0.3)     | (0.3)     | (0.4)     | (0.2)     | 1.4       | (0.5)     | 0.2       | 0.3       |
| Interest Cost Incurred              | (2.4)     | (2.3)     | (2.3)     | (1.3)     | (1.5)     | (1.8)     | (1.7)     | (1.5)     | (1.0)     | (1.1)     | (0.8)     | (1.2)     | (1.3)     |
| Interest Capitalized                | 1.5       | 1.9       | 2.1       | 1.3       | 1.5       | 1.6       | 1.1       | 0.2       | 0.8       | 0.9       | 0.7       | 0.9       | 0.9       |
| Net Income Before Taxes             | 169.7     | 171.6     | 133.1     | 93.9      | 65.2      | 86.1      | 53.6      | 7.9       | 8.3       | 13.1      | 4.5       | (2.5)     | (5.9)     |
| Provision for Income Taxes          | 5.1       | 38.8      | 33.8      | 24.2      | 18.6      | 23.3      | 7.6       | (36.6)    | 3.3       | 5.0       | 0.2       | (0.0)     | 0.0       |
| Net Income After Taxes              | 164.6     | 132.8     | 99.3      | 69.7      | 46.6      | 62.9      | 46.0      | 44.4      | 5.0       | 8.0       | 4.3       | (2.5)     | (5.9)     |

Source: Compiled by CRFA from First Solar Inc., Form 10-Q, filed May 1, 2009; Form 10-Q, filed October 31, 2008; Form 10-Q, filed July 31, 2008; Form 10-Q, filed May 2, 2008; Form 10-Q, filed November 7, 2007; Form 10-Q, filed August 2, 2007; Form 10-Q filed May 8, 2007, <http://investor.firstsolar.com/index.cfm>, accessed January 2013.

**Exhibit 16** Balance Sheet

|                                  | 28-MAR-09 | 27-DEC-08 | 27-SEP-08 | 28-JUN-08 | 29-MAR-08 | 29-DEC-07 | 29-SEP-07 | 30-JUN-07 | 31-MAR-07 | 30-DEC-06 | 30-SEP-06 |
|----------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| <b>ASSETS</b>                    |           |           |           |           |           |           |           |           |           |           |           |
| Cash and Cash Equivalents        | 624.93    | 716.22    | 581.78    | 511.24    | 590.53    | 404.26    | 117.15    | 107.80    | 325.01    | 308.09    | 31.37     |
| Money market mutual fund         |           |           |           |           |           |           |           |           |           |           |           |
| Marketable Securities            | 172.18    | 76.04     | 133.37    | 121.76    | 90.13     | 232.69    | 564.64    | 207.21    | 0.33      | 0.32      | 0.32      |
| Accounts Receivable, Gross       | 184.79    | 61.70     | 41.52     | 50.46     | 18.70     | 18.17     | 19.66     | 13.74     | 7.84      | 27.97     | 26.44     |
| Allowance for Doubtful Accounts  | 0.00      | 0.00      | (0.04)    | (0.47)    | (0.67)    | (0.01)    | 0.00      | 0.00      | 0.00      | (0.00)    | (0.00)    |
| A/R, Net                         | 184.79    | 61.70     | 41.47     | 49.99     | 18.03     | 18.17     | 19.66     | 13.74     | 7.84      | 27.97     | 26.43     |
| Accounts receivable, unbilled    | 0.00      | 0.00      | 0.00      | 0.00      | 0.00      | 0.00      | 0.00      | 0.00      | 0.00      | 0.00      | 0.00      |
| Raw Materials                    | 104.76    | 103.73    | 94.42     | 75.84     | 38.98     | 22.87     | 18.07     | 14.68     | 10.28     | 8.21      | 5.30      |
| Work in Process                  | 7.56      | 4.04      | 7.10      | 2.84      | 3.17      | 2.29      | 3.02      | 6.56      | 0.48      | 1.12      | 1.69      |
| Finished Goods                   | 19.14     | 13.79     | 24.95     | 28.22     | 16.41     | 15.04     | 10.80     | 5.61      | 4.26      | 7.18      | 3.54      |
| Inventories                      | 131.47    | 121.55    | 126.47    | 106.90    | 58.56     | 40.20     | 31.89     | 26.85     | 15.02     | 16.51     | 10.53     |
| Project assets                   |           | 0.00      |           |           |           |           |           |           |           |           |           |
| Prepaid and Other Current Assets | 102.42    | 91.25     | 66.49     | 33.08     | 35.87     | 100.66    | 58.69     | 63.68     | 39.28     | 35.63     | 21.30     |
| Deferred project costs           | 12.26     | 0.71      | 9.08      | 1.41      | 1.22      | 2.64      |           |           |           |           |           |
| Deferred Tax                     | 11.66     | 9.92      | 3.72      | 14.18     | 3.91      | 3.89      | 3.75      | 4.82      |           | 0.00      |           |
| Total Current Assets             | 1239.70   | 1077.40   | 962.38    | 838.57    | 798.25    | 802.51    | 795.79    | 424.09    | 387.49    | 388.52    | 89.95     |
| Project Assets - Non-current     |           |           |           |           |           |           |           |           |           |           |           |
| Buildings and Improvements       | 177.79    | 137.12    | 108.57    | 109.96    | 77.28     | 44.68     | 44.15     | 42.81     | 43.16     | 21.80     | 21.80     |
| Machinery and Equipment          | 685.49    | 559.57    | 436.43    | 324.07    | 301.34    | 170.13    | 161.51    | 152.66    | 153.22    | 79.80     | 77.77     |
| Office Equipment and Furniture   | 26.73     | 22.84     | 17.80     | 13.52     | 10.15     | 7.37      | 6.93      | 5.98      | 5.08      | 4.43      | 3.32      |
| Leasehold Improvements           | 14.56     | 11.50     | 11.32     | 11.08     | 4.16      | 4.05      | 3.88      | 3.09      | 3.09      | 3.09      | 2.98      |
| Land                             | 5.66      | 5.76      | 4.94      | 3.71      | 3.72      | 3.05      | 2.98      | 2.88      | 2.86      | 2.84      | 2.77      |
| Construction in Progress         | 82.72     | 206.78    | 253.42    | 278.77    | 186.76    | 243.98    | 134.45    | 68.13     | 37.53     | 85.79     | 63.53     |
| Depreciation & Amortization      | (125.29)  | (100.94)  | (82.00)   | (66.85)   | (54.02)   | (43.13)   | (36.36)   | (29.99)   | (24.02)   | (18.88)   | (15.37)   |
| Property Plant & Equip           | 867.66    | 842.62    | 750.48    | 674.27    | 529.39    | 430.10    | 317.55    | 245.56    | 220.92    | 178.87    | 156.80    |
| Deferred Tax                     | 61.99     | 61.33     | 54.85     | 55.28     | 51.58     | 51.81     | 45.07     | 34.40     |           |           |           |
| Marketable Securities            | 14.46     | 29.56     | 14.27     | 28.21     | 28.34     | 32.71     |           |           |           |           |           |
| Investment in related party      | 25.00     | 25.00     |           |           |           | 0.00      |           |           |           |           |           |
| Restricted Investment            | 30.15     | 30.06     | 29.92     | 29.95     | 27.11     | 14.70     | 14.46     | 14.02     | 8.31      | 8.22      | 6.73      |
| Inventories                      |           | 0.00      |           |           |           |           |           |           |           |           |           |
| Goodwill                         | 33.83     | 33.83     | 33.83     | 33.83     | 33.83     | 33.45     |           |           |           |           |           |
| Other Noncurrent Assets          | 15.64     | 14.71     | 14.12     | 15.02     | 8.65      | 6.03      | 4.29      | 5.14      | 2.04      | 2.65      | 1.40      |
| Intangible Assets                |           |           |           |           |           |           |           |           |           | 0.25      | 0.26      |
| Total Assets                     | 2288.43   | 2114.50   | 1859.85   | 1675.13   | 1477.16   | 1371.31   | 1177.16   | 723.21    | 618.76    | 578.51    | 255.15    |

## Exhibit 16 (continued)

|                                          | 28-MAR-09 | 27-DEC-08 | 27-SEP-08 | 28-JUN-08 | 29-MAR-08 | 29-DEC-07 | 29-SEP-07 | 30-JUN-07 | 31-MAR-07 | 30-DEC-06 | 30-SEP-06 |
|------------------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| <b>LIABILITIES</b>                       |           |           |           |           |           |           |           |           |           |           |           |
| Accounts payable                         | 41.85     | 46.25     | 36.69     | 42.48     | 21.97     | 26.44     | 14.88     | 25.39     | 11.22     | 14.00     |           |
| A/P and Accrued Expenses                 | 102.24    | 99.94     | 70.73     | 51.20     | 35.50     | 24.49     | 0.43      | 0.51      | 3.41      | 5.15      | 34.95     |
| Income tax payable                       |           |           | 0.00      | 0.00      | 0.00      | 24.47     | 23.71     | 22.41     | 20.56     | 16.34     | 6.83      |
| Short term debt                          |           |           | 26.69     | 24.63     | 17.67     | 14.84     | 9.90      | 3.33      | 3.32      | 3.31      | 2.62      |
| Current Portion of Long-term Debt        | 32.95     | 34.95     | 9.86      | 10.87     | 9.26      | 2.09      | 5.50      | 4.05      | 3.36      | 2.76      |           |
| Product warranty liability               | 5.31      | 4.04      | 24.77     | 11.77     | 9.85      | 21.86     | 14.53     | 8.16      | 3.59      | 2.64      |           |
| Accrued compensation and benefits        | 11.92     | 32.15     | 54.23     | 45.95     | 61.49     | 35.22     | 19.81     | 2.88      | 16.75     | 1.97      |           |
| Accrued property, plant and equipment    | 22.10     | 44.12     | 58.06     | 51.95     | 28.99     | 17.08     | 14.32     | 10.41     | 6.00      | 5.56      |           |
| Accrued expenses                         | 46.92     | 60.60     | 16.61     | 40.41     | 43.01     | 14.80     | 9.11      | 0.02      | 0.00      | 0.34      | 0.16      |
| Other current liabilities                | 52.54     | 59.74     |           |           |           |           |           |           |           |           | 26.00     |
| Note Payable to a Related Party          |           |           |           |           |           |           |           |           |           |           | 70.56     |
| Total Current Liabilities                | 315.84    | 381.78    | 297.63    | 279.25    | 227.74    | 181.30    | 112.19    | 77.14     | 68.19     | 52.07     |           |
| Long-term Debt                           | 195.22    | 163.52    | 140.81    | 108.55    | 70.21     | 68.86     | 72.48     | 96.48     | 71.96     | 61.05     | 35.57     |
| Accrued Recycling                        | 45.37     | 35.24     | 28.08     | 23.57     | 18.15     | 13.08     | 9.42      | 6.45      | 4.99      | 3.72      | 2.76      |
| Other Noncurrent Liabilities             | 26.34     | 20.93     | 8.16      | 13.17     | 9.88      | 10.81     | 2.24      | 1.13      | 0.31      | 0.00      | 0.06      |
| Employee Stock Options                   |           |           |           |           |           |           | 0.00      | 60.71     |           |           |           |
| Total Liabilities                        | 582.76    | 601.46    | 474.68    | 424.53    | 325.98    | 274.05    | 196.34    | 241.91    | 145.45    | 116.84    | 108.94    |
| <b>SHAREHOLDERS' EQUITY</b>              |           |           |           |           |           |           |           |           |           |           |           |
| Employee Stock Options                   |           |           |           |           |           |           |           |           | 91.10     | 50.23     | 24.94     |
| Membership Equity                        |           |           |           |           |           |           |           |           |           |           | 0.00      |
| Common Stock                             | 0.08      | 0.08      | 0.08      | 0.08      | 0.08      | 0.08      | 0.08      | 0.07      | 0.07      | 0.07      | 0.06      |
| Additional Paid-in Capital               | 1194.32   | 1176.16   | 1144.68   | 1127.25   | 1100.63   | 1079.78   | 1026.86   | 575.05    | 521.38    | 555.75    | 274.71    |
| Accumulated Deficit                      | 525.82    | 361.23    | 228.45    | 129.19    | 59.51     | 12.90     | (49.98)   | (96.01)   | (140.43)  | (145.40)  | (153.44)  |
| Other Comprehensive Income               | (14.55)   | (24.42)   | 11.96     | (5.91)    | (9.05)    | 4.52      | 3.86      | 2.20      | 1.20      | 1.02      | (0.06)    |
| Contingent consideration                 |           | 0.00      |           |           |           |           |           |           |           |           |           |
| Total Equity                             | 1705.67   | 1513.04   | 1385.17   | 1250.60   | 1151.18   | 1097.27   | 980.82    | 481.30    | 473.32    | 461.67    | 146.20    |
| Total Liabilities & Shareholders' Equity | 2288.43   | 2114.50   | 1859.85   | 1675.13   | 1477.16   | 1371.31   | 1177.16   | 723.21    | 618.76    | 578.51    | 255.15    |

Source: Compiled by CRFA from First Solar Inc., Form 10-Q, filed May 1, 2009; Form 10-Q, filed October 31, 2008; Form 10-Q, filed July 31, 2008; Form 10-Q, filed May 2, 2008; Form 10-Q, filed November 7, 2007; Form 10-Q, filed August 2, 2007; Form 10-Q filed May 8, 2007, <http://investor.firstsolar.com/index.cfm>, accessed January 2013.

## Exhibit 17 Cash Flow Quarterly

|                                          | 28-MAR-09 | 27-DEC-08 | 27-SEP-08 | 28-JUN-08 | 29-MAR-08 | 29-DEC-07 | 29-SEP-07 | 30-JUN-07 | 31-MAR-07 | 30-DEC-06 |
|------------------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| <b>OPERATING CASH FLOWS</b>              |           |           |           |           |           |           |           |           |           |           |
| Cash Receipts                            | 325.7     | 424.6     | 350.0     | 234.6     | 194.6     | 206.2     | 152.2     | 71.0      | 86.6      | 51.7      |
| Cash Payments                            | (259.7)   | (209.5)   | (211.9)   | (163.9)   | (137.8)   | (104.4)   | (53.4)    | (72.3)    | (46.4)    | (39.3)    |
| Interest received                        | 2.9       | 3.8       | 4.7       | 4.5       | 6.2       | 20.0      | -         | -         | -         | -         |
| Interest paid net of amounts capitalized | (2.2)     | (1.6)     | (1.5)     | (1.5)     | (0.0)     | (8.0)     | (0.7)     | 2.5       | 3.9       | (0.2)     |
| Excess tax benefit from share-based comp | -         | -         | -         | -         | -         | -         | -         | -         | -         | -         |
| Income tax                               | 0.7       | 1.2       | (3.8)     | (4.3)     | 4.9       | (0.8)     | 1.1       | (14.2)    | (5.0)     | -         |
| Excess tax benefit from share-based comp | (3.3)     | (14.9)    | 0.2       | (9.7)     | (4.3)     | (11.6)    | (18.6)    | -         | -         | -         |
| Other                                    | (0.3)     | (0.3)     | (0.4)     | (0.5)     | (0.3)     | (1.7)     | 0.4       | (0.5)     | (0.2)     | 1.1       |
| Cash from Operating Activities           | 63.7      | 203.3     | 137.3     | 59.2      | 63.3      | 99.7      | 80.9      | (13.6)    | 38.9      | 13.3      |
| <b>INVESTING CASH FLOWS</b>              |           |           |           |           |           |           |           |           |           |           |
| Capital Expenditure                      | (86.4)    | (128.7)   | (95.7)    | (160.3)   | (74.6)    | (96.7)    | (65.2)    | (39.6)    | (40.8)    | (55.1)    |
| Deposits                                 | -         | -         | -         | -         | -         | -         | 3.2       | (3.2)     | -         | -         |
| Purchase of Marketable Securities        | (117.6)   | (60.6)    | (106.5)   | (110.0)   | (57.8)    | (354.6)   | (487.7)   | (238.9)   | -         | -         |
| Proceeds from maturities of marketable s | 7.0       | (265.9)   | 338.6     | 23.5      | 11.3      | -         | -         | -         | -         | -         |
| Proceeds from sales of marketable securi | 29.8      | 369.3     | (229.4)   | 55.0      | 223.9     | 625.4     | 122.1     | 40.3      | -         | -         |
| Investment in notes receivable           | (13.8)    | -         | -         | -         | -         | -         | -         | -         | -         | -         |
| Payments received on notes receivables   | -         | -         | -         | -         | -         | -         | -         | -         | -         | -         |
| Increase in restricted cash              | -         | -         | -         | -         | -         | -         | -         | -         | -         | -         |
| Purchase of Restricted Investments       | (0.3)     | (0.3)     | (0.3)     | (2.9)     | (12.1)    | (0.1)     | (0.2)     | (5.7)     | (0.0)     | (1.3)     |
| Sale of (purchase of) investment in rela | -         | (25.0)    | -         | -         | -         | -         | -         | -         | -         | -         |
| Investment in project assets             | -         | -         | -         | -         | -         | -         | -         | -         | -         | -         |
| Acquisitions, net of cash acquired       | -         | -         | -         | -         | -         | (5.5)     | -         | -         | -         | -         |
| Other investing activities               | -         | -         | -         | -         | -         | -         | -         | -         | -         | -         |
| Cash from Investing Activities           | (181.2)   | (111.1)   | (93.3)    | (194.6)   | 90.7      | 168.5     | (427.8)   | (247.1)   | (40.8)    | (56.4)    |
| <b>FINANCING CASH FLOWS</b>              |           |           |           |           |           |           |           |           |           |           |
| Proceeds from stock option exercises     | 1.4       | -         | (7.2)     | 7.2       | -         | -         | 366.0     | -         | -         | 302.7     |
| Proceeds from Stock Options Exercised    | -         | 1.9       | 14.1      | (5.9)     | 5.9       | 3.4       | 3.9       | 2.2       | 0.6       | -         |
| Proceeds from borrowings under credit fa | -         | -         | -         | -         | -         | -         | -         | -         | -         | -         |
| Repayment of long term debt              | (3.9)     | (6.9)     | (4.2)     | (4.9)     | (25.7)    | (4.8)     | (28.3)    | (0.8)     | (0.8)     | (0.1)     |
| Proceeds from Debt                       | 45.3      | 44.8      | 44.6      | 49.4      | 0.1       | 1.2       | 6.9       | 26.4      | 14.8      | 34.1      |

## Exhibit 17 (continued)

|                                          | 28-MAR-09 | 27-DEC-08 | 27-SEP-08 | 28-JUN-08 | 29-MAR-08 | 29-DEC-07 | 29-SEP-07 | 30-JUN-07 | 31-MAR-07 | 30-DEC-06 |
|------------------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| Proceeds from Note Payable               | -         | -         | -         | -         | -         | -         | -         | -         | -         | -         |
| Repayment of notes payable to a related  | -         | -         | -         | -         | -         | -         | -         | -         | -         | (26.0)    |
| Equity Contributions                     | -         | -         | -         | -         | -         | -         | -         | -         | -         | -         |
| Excess tax benefit from share-based comp | 3.3       | -         | -         | (4.3)     | 4.3       | -         | -         | (0.1)     | 0.1       | -         |
| Excess tax benefit from share-based comp | -         | 14.9      | (0.2)     | 14.0      | -         | 11.6      | 4.6       | 14.0      | -         | -         |
| Debt Issuance Costs                      | -         | -         | -         | -         | -         | -         | -         | -         | -         | -         |
| Proceeds from economic development fundi | 0.6       | -         | -         | -         | 35.7      | 2.9       | 1.8       | 0.8       | 4.0       | 8.7       |
| Repurchase of Membership Units           | -         | -         | -         | -         | -         | -         | -         | -         | -         | -         |
| Other Financing Activities               | (0.0)     | -         | (0.0)     | (0.0)     | (0.0)     | -         | (0.0)     | -         | (0.0)     | (0.0)     |
| Cash from Financing Activities           | 46.7      | 54.8      | 47.2      | 55.4      | 20.2      | 14.3      | 354.8     | 42.6      | 18.7      | 319.3     |

Source: Compiled by CRFA from First Solar Inc., Form 10-Q, filed May 1, 2009; Form 10-Q, filed October 31, 2008; Form 10-Q, filed July 31, 2008; Form 10-Q, filed May 2, 2008; Form 10-Q, filed November 7, 2007; Form 10-Q, filed August 2, 2007; Form 10-Q filed May 8, 2007, <http://investor.firstsolar.com/index.cfm>, accessed January 2013.

## Endnotes

<sup>1</sup> Dan Mahoney, interviewed by the authors, New York, NY (by phone), October 10, 2012.

<sup>2</sup> Dan Mahoney, interviewed by the authors, New York, NY (by phone), October 10, 2012.

<sup>3</sup> Dan Mahoney, interviewed by the authors, New York, NY (by phone), October 10, 2012.

<sup>4</sup> Dan Mahoney, interviewed by the authors, New York, NY (by phone), October 10, 2012.

<sup>5</sup> Tony Andersson, "Solar Energy Investment Theme: It's Always Darkest Before Dawn," UBS Wealth Management Research, March 5, 2009.

<sup>6</sup> Robert Kroop, "Solar Expected to Maintain its Status as the World's Fastest-Growing Energy Technology," *Sustainability Investment News*, March 3, 2009, <http://www.socialfunds.com/news/article.cgi/2639.html>, accessed October 2012.

<sup>7</sup> Robert Kroop, "Solar Expected to Maintain its Status as the World's Fastest-Growing Energy Technology," *Sustainability Investment News*, March 3, 2009, <http://www.socialfunds.com/news/article.cgi/2639.html>, accessed October 2012.

<sup>8</sup> Tony Andersson, "Solar Energy Investment Theme: It's Always Darkest Before Dawn," UBS Wealth Management Research, March 5, 2009.

<sup>9</sup> Tony Andersson, "Solar Energy Investment Theme: It's Always Darkest Before Dawn," UBS Wealth Management Research, March 5, 2009.

<sup>10</sup> Tony Andersson, "Solar Energy Investment Theme: It's Always Darkest Before Dawn," UBS Wealth Management Research, March 5, 2009.

<sup>11</sup> Tony Andersson, "Solar Energy Investment Theme: It's Always Darkest Before Dawn," UBS Wealth Management Research, March 5, 2009.

<sup>12</sup> Michaela D. Platzer, "U.S. Solar Photovoltaic Manufacturing: Industry trends, Global Competition, Federal Support," *Congressional Research Service*, June 13, 2012, <http://www.fas.org/sgp/crs/misc/R42509.pdf>, accessed November 2012.

<sup>13</sup> Morgan Jerome Hallmon, Robert Siegel and Robert Burgelman, "First Solar, Inc. in 2010," Stanford Graduate School of Business case no. SM-190 (Stanford Graduate School of Business, October 2010), via Harvard Business Online, <http://hbsp.harvard.edu/>, accessed October 2012.

<sup>14</sup> Morgan Jerome Hallmon, Robert Siegel and Robert Burgelman, "First Solar, Inc. in 2010," Stanford Graduate School of Business case no. SM-190 (Stanford Graduate School of Business, October 2010), via Harvard Business Online, <http://hbsp.harvard.edu/>, accessed October 2012.

<sup>15</sup> Tony Andersson, "Solar Energy Investment Theme: It's Always Darkest Before Dawn," UBS Wealth Management Research, March 5, 2009.

<sup>16</sup> Green Rhino Energy website, "PV Modules Overview," [http://www.greenrhinoenergy.com/solar/technologies/pv\\_modules.php](http://www.greenrhinoenergy.com/solar/technologies/pv_modules.php), accessed November 2012.

<sup>17</sup> National Geographic website, "Green Living: Solar Panel Energy vs. Size," <http://greenliving.nationalgeographic.com/solar-panel-energy-vs-size-2438.html>, accessed November 2012.

<sup>18</sup> Michaela D. Platzer, "U.S. Solar Photovoltaic Manufacturing: Industry trends, Global Competition, Federal Support," *Congressional Research Service*, June 13, 2012, <http://www.fas.org/sgp/crs/misc/R42509.pdf>, accessed November 2012.

<sup>19</sup> Michaela D. Platzer, "U.S. Solar Photovoltaic Manufacturing: Industry trends, Global Competition, Federal Support," *Congressional Research Service*, June 13, 2012, <http://www.fas.org/sgp/crs/misc/R42509.pdf>, accessed November 2012.

<sup>20</sup> Tony Andersson, "Solar Energy Investment Theme: It's Always Darkest Before Dawn," UBS Wealth Management Research, March 5, 2009.

<sup>21</sup> Rocky Mountain Institute website, "Solar PV Balance of System," <http://www.rmi.org/SolarPVBOS>, accessed November 2012.

<sup>22</sup> Morgan Jerome Hallmon, Robert Siegel and Robert Burgelman, "First Solar, Inc. in 2010," Stanford Graduate School of Business case no. SM-190 (Stanford Graduate School of Business, October 2010), via Harvard Business Online, <http://hbsp.harvard.edu/>, accessed October 2012.

<sup>23</sup> Morgan Jerome Hallmon, Robert Siegel and Robert Burgelman, "First Solar, Inc. in 2010," Stanford Graduate School of Business case no. SM-190 (Stanford Graduate School of Business, October 2010), via Harvard Business Online, <http://hbsp.harvard.edu/>, accessed October 2012.

<sup>24</sup> "First Solar Announces Acquisition of Turner Renewable Energy," First Solar press release, November 30, 2007, on First Solar website, <http://investor.firstsolar.com/releasedetail.cfm?releaseid=279341>, accessed October 2012.

<sup>25</sup> Nicholas Groom, "First Solar Buys Rival Utility Project Pipeline," *Reuters*, March 3, 2009, <http://www.reuters.com/article/2009/03/03/us-firstsolar-optisolar-idUSTRE5225W420090303>, accessed October 2012.

<sup>26</sup> Morgan Jerome Hallmon, Robert Siegel and Robert Burgelman, "First Solar, Inc. in 2010," Stanford Graduate School of Business case no. SM-190 (Stanford Graduate School of Business, October 2010), via Harvard Business Online, <http://hbsp.harvard.edu/>, accessed October 2012.

<sup>27</sup> First Solar website, "Completed Projects," <http://www.firstsolar.com/Projects/Completed-Projects>, accessed October 2012.

<sup>28</sup> First Solar, 2009 Annual Report, p. 39, <http://investor.firstsolar.com/secfiling.cfm?filingID=1274494-10-7>, accessed October 2012.

<sup>29</sup> Sean Harrington and Dennis Rohan, "Note on Energy," Stanford Graduate School of Business case no. E-302 (Stanford Graduate School of Business, February 2008), via Harvard Business Online, <http://hbsp.harvard.edu/>, accessed October 2012.

<sup>30</sup> Maja Wessels (Executive VP), "Presentation on Thin Film PV," First Solar PowerPoint presentation, December 2009, [http://www.caissedesdepots.fr/fileadmin/PDF/international/first\\_solar\\_presentation.pdf](http://www.caissedesdepots.fr/fileadmin/PDF/international/first_solar_presentation.pdf), accessed October 2012.

<sup>31</sup> Maja Wessels (Executive VP), "Presentation on Thin Film PV," First Solar PowerPoint presentation, December 2009, [http://www.caissedesdepots.fr/fileadmin/PDF/international/first\\_solar\\_presentation.pdf](http://www.caissedesdepots.fr/fileadmin/PDF/international/first_solar_presentation.pdf), accessed October 2012.

<sup>32</sup> Tony Andersson, "Solar Energy Investment Theme: It's Always Darkest Before Dawn," UBS Wealth Management Research, March 5, 2009.

<sup>33</sup> Tony Andersson, "Solar Energy Investment Theme: It's Always Darkest Before Dawn," UBS Wealth Management Research, March 5, 2009.

<sup>34</sup> David Edwards, "First Solar (FSLR, \$134.95, Overweight-V, Price Target \$175)," Morgan Stanley Research, October 16, 2007.

<sup>35</sup> David Edwards, "First Solar (FSLR, \$134.95, Overweight-V, Price Target \$175)," Morgan Stanley Research, October 16, 2007.

<sup>36</sup> Steve O'Rourke, Hari Polavarapu & Peter Kim, "First Solar Inc.: Addressing Some Investor Concerns...", Deutsche Bank, June 4, 2008.

<sup>37</sup> Steve O'Rourke, Hari Polavarapu & Peter Kim, "First Solar Inc.: Solid Execution, Upside to Estimates," Deutsche Bank, July 30, 2008.

<sup>38</sup> Satya Kumar & Viswanath Valluri, "First Solar: C2Q08 Preview," Credit Suisse, July 13, 2008.

<sup>39</sup> David Edwards, "First Solar (FSLR, \$134.95, Overweight-V, Price Target \$175)," Morgan Stanley Research, October 16, 2007.

<sup>40</sup> Satya Kumar & Viswanath Valluri, "First Solar: C2Q08 Preview," Credit Suisse, July 13, 2008.

<sup>41</sup> David Edwards, "First Solar (FSLR, \$134.95, Overweight-V, Price Target \$175)," Morgan Stanley Research, October 16, 2007.

<sup>42</sup> David Edwards, "First Solar (FSLR, \$134.95, Overweight-V, Price Target \$175)," Morgan Stanley Research, October 16, 2007.

<sup>43</sup> Paul Clegg, David Paek & Patrick Uotila, "First Solar, Initiating Coverage: An "All Weather" Solar Story" December 19, 2008.

<sup>44</sup> Steve O'Rourke, Hari Polavarapu & Peter Kim, "First Solar Inc.: Addressing Some Investor Concerns...", Deutsche Bank, June 4, 2008.

<sup>45</sup> Satya Kumar & Viswanath Valluri, "First Solar: Trimming Target Price to Reflect Currency Risk," Credit Suisse, September 9, 2008.

<sup>46</sup> Satya Kumar & Viswanath Valluri, "First Solar: Trimming Target Price to Reflect Currency Risk," Credit Suisse, September 9, 2008.

<sup>47</sup> Dan Mahoney, Jean-Louis Thiémélé, Jon Dye, "Diversified Industrials – Solar Energy," CFRA Industry Report, January 11, 2008.

<sup>48</sup> Dan Mahoney, Jean-Louis Thiémélé, Jon Dye, "Diversified Industrials – Solar Energy," CFRA Industry Report, January 11, 2008.

<sup>49</sup> Tony Andersson, "Solar Energy Investment Theme: It's Always Darkest Before Dawn," UBS Wealth Management Research, March 5, 2009.

<sup>50</sup> Dan Mahoney, "Diversified Industrials – Solar Energy," CFRA Industry Report, July 22, 2008.

<sup>51</sup> Tony Andersson, "Solar Energy Investment Theme: It's Always Darkest Before Dawn," UBS Wealth Management Research, March 5, 2009.

<sup>52</sup> Tony Andersson, "Solar Energy Investment Theme: It's Always Darkest Before Dawn," UBS Wealth Management Research, March 5, 2009.

<sup>53</sup> Dino Sadikovic, "First Solar Inc.," CFRA Company Report, June 10, 2009.

<sup>54</sup> Dino Sadikovic, "First Solar Inc.," CFRA Company Report, June 10, 2009.

<sup>55</sup> Satya Kumar & Viswanath Valluri, "First Solar: Charting its Own Course," Credit Suisse, April 30, 2009.

<sup>56</sup> Satya Kumar & Viswanath Valluri, "First Solar: Charting its Own Course," Credit Suisse, April 30, 2009.

<sup>57</sup> Steve O'Rourke, Hari Polavarapu & Peter Kim, "Solid Execution in a Tough Environment," Deutsche Bank, April 29, 2009.

<sup>58</sup> Steve O'Rourke, Hari Polavarapu & Peter Kim, "Solid Execution in a Tough Environment," Deutsche Bank, April 29, 2009.

<sup>59</sup> Paul Clegg, David Paek and Patrick Uotila, "First Solar: Raising PT to \$215 - Cost Continues to Drop," Jefferies and Company, April 30, 2009.

<sup>60</sup> Paul Clegg, David Paek and Patrick Uotila, "First Solar: Raising PT to \$215 - Cost Continues to Drop," Jefferies and Company, April 30, 2009.

<sup>61</sup> Paul Clegg, David Paek and Patrick Uotila, "First Solar: Raising PT to \$215 - Cost Continues to Drop," Jefferies and Company, April 30, 2009.

<sup>62</sup> Dino Sadikovic, "First Solar Inc.," CFRA Company Report, June 10, 2009.

<sup>63</sup> Dino Sadikovic, "First Solar Inc.," CFRA Company Report, June 10, 2009.

<sup>64</sup> Dino Sadikovic, "First Solar Inc.," CFRA Company Report, June 10, 2009.

<sup>65</sup> Dino Sadikovic, "First Solar Inc.," CFRA Company Report, June 10, 2009.

<sup>66</sup> Dino Sadikovic, "First Solar Inc.," CFRA Company Report, June 10, 2009.